

Type "HXXX" Faults

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H001:	H001:COILFAULT PLAT UP1:BAT-	Short circuit of the platform up #1 circuit to battery negative.	• Short circuit in platform up #1 harness. • Platform up #1 coil short circuit. • GCON ECM.	Platform up function inhibited.
H002:	H002:COILFAULT PLAT UP1:OPEN	Open circuit in the platform up #1 circuit.	• Short circuit in platform up #1 harness. • Platform up #1 coil open circuit. • GCON ECM.	Platform up function inhibited.
H003:	H003:COILFAULT PLAT UP1:BAT+	Short circuit of the platform up #1 circuit to battery positive.	• Short circuit in platform up #1 harness. • Platform up #1 coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H009:	H009:COILFAULT PLAT DOWN1:BAT+	Short circuit of the platform down #1 circuit to battery positive.	• Short circuit in platform down #1 harness. • Platform down #1 coil short circuit. • GCON ECM.	All functions inhibited.
H013:	H013:COILFAULT DRIVE FWD1:BAT-	Short circuit of the drive forward #1 circuit to battery negative.	• Short circuit in drive forward #1 harness. • Drive forward #1 coil short circuit. • GCON ECM.	Drive forward function inhibited.
H014:	H014:COILFAULT DRIVE FWD1:OPEN	Open circuit in the drive forward #1 circuit.	• Open circuit in drive forward #1 harness. • Drive forward #1 coil open circuit. • GCON ECM.	Drive forward function inhibited.
H015:	H015:COILFAULT DRIVE FWD1:BAT+	Short circuit of the drive forward #1 circuit to battery positive.	• Short circuit in drive forward #1 harness. • Drive forward #1 coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H019:	H019:COILFAULT DRIVE REV1:BAT-	Short circuit of the drive reverse #1 circuit to battery negative.	• Short circuit in drive reverse #1 harness. • Drive reverse #1 coil short circuit. • GCON ECM.	Drive reverse function inhibited.
H020:	H020:COILFAULT DRIVE REV1:OPEN	Open circuit in the drive reverse #1 circuit.	• Open circuit in drive reverse #1 harness. • Drive reverse #1 coil open circuit. • GCON ECM.	Drive reverse function inhibited.
H021:	H021:COILFAULT DRIVE REV1:BAT+	Short circuit of the drive reverse #1 circuit to battery positive.	• Short circuit in drive reverse #1 harness. • Drive reverse #1 coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H027:	H027:COILFAULT DRIVE STEER RIGHT:BAT+	Short circuit of the steer right circuit to battery positive.	• Short circuit in steer right harness. • Steer right coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H030:	H030:COILFAULT DRIVE STEER LEFT:BAT+	Short circuit of the steer left circuit to battery positive.	• Short circuit in steer left harness. • Steer left coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H033:	H033:COILFAULT HI/LO SPEED:BAT+	Short circuit of the hi/lo speed coil to battery positive.	• Short circuit in Hi/Lo speed coil harness. • Hi/Lo speed coil short circuit. • GCON ECM.	Hi/Lo speed coil disabled. Machine will operate in low speed mode.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H043:	H043:COILFAULT BRAKE REL:BAT-	Short circuit of the brake circuit to battery negative.	• Short circuit in brake release enable harness. • Brake release relay short circuit. • GCON ECM.	All functions inhibited except platform down.
H045:	H045:COILFAULT BRAKE REL:BAT+	Short circuit of the brake circuit to battery positive.	• Short circuit in brake release enable harness. • Brake release relay short circuit. • GCON ECM.	All functions inhibited except platform down.
H049:	H049:COILFAULT O/R EXTEND:BAT-	Short circuit of the outrigger extend coil to battery negative.	• Short circuit in outrigger extend coil harness. • Outrigger extend coil short circuit. • GCON ECM.	Only outrigger extend function disabled.
H050:	H050:COILFAULT O/R EXTEND:OPEN	Open circuit in the outrigger extend coil circuit.	• Short circuit in outrigger extend coil harness. • Outrigger extend coil open circuit. • GCON ECM.	Only outrigger extend function disabled.
H051:	H051:COILFAULT O/R EXTEND:BAT+	Short circuit of the outrigger extend coil to battery positive.	• Short circuit in outrigger extend coil harness. • Outrigger extend coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H052:	H052:COILFAULT O/R RETRACT:BAT-	Short circuit of the outrigger retract coil to battery negative.	• Short circuit in outrigger retract coil harness. • Outrigger retract coil short circuit. • GCON ECM.	Only outrigger retract function disabled.
H053:	H053:COILFAULT O/R RETRACT:OPEN	Open circuit in the outrigger retract coil circuit.	• Short circuit in outrigger retract coil harness. • Outrigger retract coil open circuit. • GCON ECM.	Only outrigger retract function disabled.
H054:	H054:COILFAULT O/R RETRACT:BAT+	Short circuit of the outrigger retract coil to battery positive.	• Short circuit in outrigger retract coil harness. • Outrigger retract coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H057:	H057:COILFAULT LF RIGGER:BAT+	Short circuit of the left front outrigger coil to battery positive.	• Short circuit in left front outrigger coil harness. • Left front outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H060:	H060:COILFAULT LR RIGGER:BAT+	Short circuit of the left rear outrigger coil to battery positive.	• Short circuit in left rear outrigger coil harness. • Left rear outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H063:	H063:COILFAULT RF RIGGER:BAT+	Short circuit of the right front outrigger coil to battery positive.	• Short circuit in right front outrigger coil harness. • Right front outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H066:	H066:COILFAULT RR RIGGER:BAT+	Short circuit of the right rear outrigger coil to battery positive.	• Short circuit in right rear outrigger coil harness. • Right rear outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H067:	H067:FAULT MC ENABLE:BAT-	Short circuit of the motor controller circuit to battery negative.	• Short circuit in motor controller enable wire. • Motor Controller. • Contactor coil shorted to battery negative. • GCON ECM.	All functions inhibited except platform down.
H069:	H069:FAULT MC ENABLE:BAT+	Short circuit of the motor controller circuit to battery positive.	• Short circuit in motor controller enable wire. • Motor Controller. • Contactor Coil. • GCON ECM.	All functions inhibited except platform down.
H070:	H070:FAULT MC THROTTLE:BAT-	Short circuit of the motor controller throttle circuit to battery negative.	• Short circuit in motor controller throttle wire. • Motor Controller. • GCON ECM.	All functions inhibited except platform down.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H072:	H072:FAULT MC THROTTLE:BAT+	Short circuit of the motor controller throttle circuit to battery positive.	• Short circuit in motor controller throttle wire. • Motor Controller. • GCON ECM.	All functions inhibited except platform down.
H074:	H074:COILFAULT LF RIGGER	Short circuit of the left front outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in left front outrigger harness. • Left front outrigger coil short or open circuit. • GCON ECM.	Left front outrigger function inhibited.
H075:	H075:COILFAULT LR RIGGER	Short circuit of the left rear outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in left rear outrigger harness. • Left rear outrigger coil short or open circuit. • GCON ECM.	Left rear outrigger function inhibited.
H076:	H076:COILFAULT RF RIGGER	Short circuit of the right front outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in right front outrigger harness. • Right front outrigger coil short or open circuit. • GCON ECM.	Right front outrigger function inhibited.
H077:	H077:COILFAULT RR RIGGER	Short circuit of the right rear outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in right rear outrigger harness. • Right rear outrigger coil short or open circuit. • GCON ECM.	Right rear outrigger function inhibited.
H078:	H078:COILFAULT PLAT DOWN1	Short circuit of the platform down #1 circuit to battery positive/negative or open circuit.	• Short or open circuit in platform down #1 harness. • Platform down #1 coil short or open circuit. • GCON ECM.	Platform down function inhibited.
H079:	H079:COILFAULT HI/LO SPEED	Short circuit of the HI/LO speed circuit to battery positive/negative or open circuit.	• Short circuit in HI/LO speed harness. • HI/LO speed coil short or open circuit. • GCON ECM.	Hi/Lo speed coil disabled. Machine will operate in high speed mode if open circuit or short with battery negative. Machine will operate in low speed mode if short with battery positive.
H080:	H080:COILFAULT STEER LEFT	Short circuit of the steer left circuit to battery negative or open circuit.	• Short or open circuit in steer left harness. • Steer left coil short or open circuit. • GCON ECM.	Steer left function inhibited.
H081:	H081:COILFAULT STEER RIGHT	Short circuit of the steer right circuit to battery negative or open circuit.	• Short or open circuit in steer right harness. • Steer right coil short or open circuit. • GCON ECM.	Steer right function inhibited.
H102:	H102:COILFAULT RELIEF VALVE:BAT+	Short circuit of the relief valve circuit to battery positive.	• Short circuit in relief valve harness. • Relief valve coil short circuit. • GCON ECM.	All functions inhibited except platform down while above limit switch.
H103:	H103:COILFAULT RELIEF VALVE:BAT-	Short circuit of the relief valve circuit to battery negative.	• Short circuit in relief valve harness. • Relief valve coil short circuit. • GCON ECM.	All functions inhibited except platform down while above limit switch.
H104:	H104:COILFAULT RELIEF VALVE:OPEN	Relief valve circuit open.	• Open circuit in relief valve harness. • Relief valve coil open circuit. • GCON ECM.	All functions inhibited except platform down while above limit switch.

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Type "HXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H106:	H106:COILFAULT DECK EXTEND:BAT-	Short circuit of the platform extend circuit to battery negative.	• Short circuit in PED harness. • PED extend valve coil short circuit. • GCON ECM.	All functions inhibited.
H107:	H107:COILFAULT DECK EXTEND:OPEN	Open circuit in the platform extend circuit.	• Open circuit in PED harness. • PED extend valve coil open circuit. • GCON ECM.	All functions inhibited.
H108:	H108:COILFAULT DECK EXTEND:BAT+	Short circuit of the platform extend circuit to battery positive.	• Short circuit in PED harness. • PED extend valve coil short circuit. • GCON ECM.	All functions inhibited.
H109:	H109:COILFAULT DECK RETRACT:BAT-	Short circuit of the platform retract circuit to battery negative.	• Short circuit in PED harness. • PED retract valve coil short circuit. • GCON ECM.	All functions inhibited.
H110:	H110:COILFAULT DECK RETRACT:OPEN	Open circuit in the platform retract circuit.	• Open circuit in PED harness. • PED retract valve coil open circuit. • GCON ECM.	All functions inhibited.
H111:	H111:COILFAULT DECK RETRACT:BAT+	Short circuit of the platform retract circuit to battery positive.	• Short circuit in PED harness. • PED retract valve coil short circuit. • GCON ECM.	All functions inhibited.
H114:	H114:COILFAULT DECK ENABLE:BAT+	Short circuit of the platform enable circuit to battery positive.	• Short circuit in PED harness. • PED enable valve coil short circuit. • GCON ECM.	All functions inhibited.
H115:	H115:COILFAULT DECK ENABLE	Short circuit of the platform enable circuit.	• Short circuit in PED harness. • GCON ECM.	All functions inhibited.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
P001:	P001:PWR FAULT SW PWR1:BAT-	Short circuit of the switched power #1 circuit to battery negative.	• Short circuit in switched power #1, down limit switch, pothole limit switch, digital tilt switch harness. • GCON ECM.	All functions inhibited.
P003:	P003:PWR FAULT SW PWR1:BAT+	Short circuit of the switched power #1 circuit to battery positive.	• Short circuit in switched power #1, down limit switch, pothole limit switch, digital tilt switch harness. • GCON ECM.	All functions inhibited.
P004:	P004:DEVICEFAULT HORN:BAT-	Short circuit of the automotive horn circuit to battery negative.	• Short circuit in automotive horn harness. • Automotive horn short circuit. • GCON ECM.	Automotive horn inhibited.
P005:	P005:DEVICEFAULT HORN:OPEN	Open circuit of the automotive horn circuit.	• Open circuit in automotive horn harness. • Automotive horn open circuit. • GCON ECM.	Automotive horn inhibited.
P006:	P006:DEVICEFAULT HORN:BAT+	Short circuit of the automotive horn circuit to battery positive.	• Short circuit in automotive horn harness. • Automotive horn short circuit. • GCON ECM.	Automotive horn inhibited.
P007:	P007:DEVICEFAULT GCON ALARM:BAT-	Short circuit of the GCON alarm circuit to battery negative.	• Short circuit in GCON alarm harness. • GCON alarm short circuit. • GCON ECM.	GCON alarm inhibited.
P009:	P009:DEVICEFAULT GCON ALARM:BAT+	Short circuit of the GCON alarm circuit to battery positive.	• Short circuit in GCON alarm harness. • GCON alarm short circuit. • GCON ECM.	GCON alarm inhibited.
P013:	P013:PWR FAULT PCON PWRET:BAT-	Short circuit of the PCON power return circuit to battery negative.	• Short circuit in PCON power return harness. • GCON ECM.	All functions inhibited.
P015:	P015:PWR FAULT PCON PWRET:BAT+	Short circuit of the PCON power return circuit to battery positive.	• Short circuit in PCON power return harness. • GCON ECM.	All functions inhibited.
P018:	018:PWR FAULT PCON POWER:BAT-	Short circuit of the PCON power circuit to battery negative.	• Short circuit in PCON power harness. • GCON ECM.	All functions inhibited.
P019:	018:PWR FAULT PCON POWER:BAT+	Short circuit of the PCON power circuit to battery positive.	• Short circuit in PCON power harness. • GCON ECM.	All functions inhibited.

Type "PXXX" Faults

Type "PXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
P023:	P023:PUMP MOTOR VOLTAGE NOT OK	Pump motor voltage out of range.	• Pump voltage to low. • Pump voltage to high. • DCON ECM.	All functions inhibited.
P025:	P025:PUMP MOTOR CURRENT FEEDBK+	Open or short circuit of the P-circuit from the right rear drive controller to the pump motor.	• Open or short circuit in pump motor P-cable. • Faulty pump. • DCON ECM.	All functions inhibited.
P026:	P026:MOTOR RIGHT REAR VOLTAGE NOT OK	Open or short circuit of the U or W circuit from the right rear drive controller to the right rear drive motor.	• Open circuit. in right rear drive motor U or W cable. • Drive motor. • DCON ECM.	All functions inhibited.
P028:	P028:CONTACTOR STUCK CLOSED	Main contactor stuck in the closed position.	• Short circuit in main contactor harness. • Faulty contactor. • DCON ECM.	All functions inhibited.
P029:	P029:CONTACTOR DOES NOT CLOSE	Main contactor stuck in the open position.	• Open circuit in main contactor harness. • Faulty contactor. • DCON ECM.	All functions inhibited.
P030:	P030:COIL FAULT	Short circuit of the main contactor coil to battery negative or open circuit.	• Open or short circuit in main contactor harness. • Faulty contactor. • DCON ECM.	All functions inhibited.
P037:	P037:BATTERY OUT OF RANGE	Battery voltage is out of range at startup.	• Open or short circuit in voltage sensor circuits. • Low batteries. • Faulty battery. • Battery charger connected to AC power source. • DCON ECM.	All functions inhibited.
P039:	P039:DEVICE FAULT BRAKE PWR RELAY	Open or short circuit of the brake relay contact or coil.	• Open or short circuit in brake relay harness. • Brake relay contact stuck closed. • Faulty brake relay. • GCON ECM.	All functions inhibited.
P040:	P040:COIL FAULT BRAKE PWR OPEN	Open or short circuit of the brake relay contact or coil to battery negative.	• Open or short circuit in brake relay harness. • Faulty brake relay. • Left or right DCON ECM. • GCON ECM.	All functions inhibited.

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Type "PXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
P053:	P053:COIL FAULT BRAKE/LC:Bat+	Open or short circuit of the brake circuit to battery positive.	• Open or short on driver load. • Faulty connections or harness. • DCON ECM.	All functions inhibited except platform down.
P054:	P054:COIL FAULT MC:OPEN	Open or short circuit of the motor controller.	• Open or short on driver load. • Faulty connections or harness. • DCON ECM.	All functions inhibited except platform down.
P056:	P056:MOTOR CURRENT FEEDBACK	Short circuit of the motor controller circuit to battery negative.	• Leakage to chassis frame from phase U, V or W. • Drive motor. • DCON ECM.	All functions inhibited except platform down.
P057:	P057:MOTOR CURRENT OVERLOAD	Short circuit of the drive motor circuit.	• External short of U, V or W motor connection. • Speed encoder noise. • Motor parameters out of range. • DCON ECM.	All functions inhibited except platform down.
P059:	P059:MOTOR RF PHASE OPEN	Open circuit of the right front drive motor.	• Open circuit. • Faulty connections or harness.	All functions inhibited except platform down.
P060:	P060:MOTOR LF PHASE OPEN	Open circuit of the left front drive motor.	• Open circuit. • Faulty connections or harness.	All functions inhibited except platform down.
P061:	P061:COIL FAULT SUPPLY: Bat-	DCON coil supply fault	• Coil supply shorted.	All functions inhibited except platform down.
P070:	P070:COIL FAULT BRAKE RF:B+	Short circuit of the right front brake coil to battery positive	• Open or short on driver load. • Faulty connector pins at controller or contactor coil. • Faulty connections or harness.	All functions inhibited except platform down.
P071:	P071:COIL FAULT BRAKE LF:B+	Short circuit of the left front brake coil to battery positive	• Open or short on driver load. • Faulty connector pins at controller or contactor coil. • Faulty connections or harness.	All functions inhibited except platform down.
P072:	P072:COIL FAULT BRAKE RF:B-	Short circuit of the right front brake coil to battery negative	• Open or short on driver load. • Faulty connector pins at controller or contactor coil. • Faulty connections or harness.	All functions inhibited except platform down.
P073:	P073:COIL FAULT BRAKE LF:B-	Short circuit of the left front brake coil to battery negative	• Open or short on driver load. • Faulty connector pins at controller or contactor coil. • Faulty connections or harness.	All functions inhibited except platform down.

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Type "PXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
P074:	P074:BATTERY CURRENT TOO HIGH	Battery current limit exceeded	- Function has drawn too much current from the battery - Excessive charge current while battery temperature is below freezing	All functions inhibited except platform down.
P075:	P075:BATTERY SHORTED TO 0V	Short circuit of battery positive to battery negative	- Short circuit in battery cables - Battery charger - DCON ECM.	All functions inhibited except platform down.
P076:	P076:BATTERY VOLTAGE TOO HIGH	Battery overvoltage	- Braking charge current - Battery charger	All functions inhibited except platform down.
P077:	P077:BATTERY VOLTAGE TOO LOW	Battery undervoltage	- Storage without putting battery in storage mode - High current draw while battery temperature is below freezing	All functions inhibited except platform down.
P078:	P078:RELIEF COIL: SHORTED TO SUPPLY	Relief coil shorted to battery positive	- Motor controller connector 2, pin 2 is open circuit - Motor controller connector 2, pin 2 is shorted to supply	All functions inhibited except platform down.
P079:	P079:RELIEF COIL SHORTED TO 0V	Relief coil shorted to battery negative	- Motor controller connector 2, pin 2 is shorted to supply - Overcurrent of motor controller connector 2, pin 2	All functions inhibited except platform down.
P080:	P080:MOTOR SHORT ACTIVE	Motor controller short detected	DCON ECM.	All functions inhibited except platform down.
P081:	P081:BRAKE INPUT FAULT	Brake input fault detected	DCON ECM.	All functions inhibited except platform down.
P082:	P082:PUMP HDP SHORTED SUPPLY	Pump Throttle Shorted to Supply	DCON ECM.	All functions inhibited except platform down.
P083:	P083:PUMP BDI TOO LOW	Pump voltage too low	Charge batteries	All functions inhibited except platform down.
P084:	P084:COIL SUPPLY FAULT	Coil supply fault	- Short on driver loads - Dirty connector pins at device - DCON ECM.	All functions inhibited except platform down.
P085:	P085:BMS CUTBACK FAULT	BMS cutback fault	DCON ECM.	All functions inhibited except platform down.
P086:	P086:BATTERY DMO CTRL:Bat+	DMO control short to battery positive	Short circuit in battery cables	All functions inhibited.
P087:	P087:BATTERY DMO CTRL:Bat-	DMO control short to battery negative	Short circuit in battery cables	All functions inhibited.

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Type "PXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
P088:	P088:LIFT MOTOR CURRENT OVERLOAD	Short circuit of the lift motor circuit.	• External short of U, V or W motor connection. • Speed encoder noise. • Motor parameters out of range. • DCON ECM.	All functions inhibited except platform down.
P089:	P089: COIL FAULT BRAKE LIFT:Bat+	Short circuit of the lift brake coil to battery positive.	• Open or short on driver load. • Faulty connections or harness. • DCON ECM.	All functions inhibited except platform down.
P090:	P090:COIL FAULT BRAKE LIFT:Bat-	Short circuit of the lift brake coil to battery negative.	• Open or short on driver load. • Faulty connections or harness. • DCON ECM.	All functions inhibited except platform down.
P091:	P091:LIFT MOTOR PHASE OPEN	Open circuit of the lift motor.	• Open circuit. • Faulty connections or harness.	All functions inhibited except platform down.
P092:	P092:LIFT MOTOR SHORT ACTIVE	Motor controller short detected.	• DCON ECM.	All functions inhibited except platform down.
P093:	P093:LIFT MOTOR CURRENT FEEDBACK	Short circuit of the lift motor controller circuit to battery negative.	• Leakage to chassis frame from phase U, V or W. • Lift motor. • DCON ECM.	All functions inhibited except platform down.
P094:	P094:LED FAULT BEACON:Bat-	Short circuit of the beacon LED to battery negative.	• Faulty connections or harness.	All functions inhibited except platform down.
P095:	P095:LED FAULT BEACON:Bat+	Short circuit of the beacon LED to battery positive.	• Faulty connections or harness.	All functions inhibited except platform down.
P096:	P096:LCON 12V SUPPLY FAILURE	LCON 12V supply failed.	• Coil supply shorted.	All functions inhibited except platform down.
P097:	P097: DCON DRIVER 1 FAULT	DCON driver 1 failed.	• DCON ECM.	All functions inhibited except platform down.
P098:	P098: DCON DRIVER 5 FAULT	DCON driver 5 failed.	• DCON ECM.	All functions inhibited except platform down.
P099:	P099:LIFT BRAKE INPUT FAULT	Input to the lift brake failed.	• DCON ECM.	All functions inhibited except platform down.
P100:	P100:LCON COIL SUPPLY FAULT	Coil supply failed.	• Coil supply shorted.	All functions inhibited except platform down.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
P101:	P101:CONTACTOR STUCK CLOSED	Main contactor stuck in the closed position.	• Short circuit in main contactor harness. • Faulty contactor. • DCON ECM.	All functions inhibited.
P102:	P102:CONTACTOR DOES NOT CLOSE	Main contactor stuck in the open position.	• Open circuit in main contactor harness. • Faulty contactor. • DCON ECM.	All functions inhibited.
P103:	P103: LCON DRIVER 1 FAULT	LCON driver 1 failed.	• LCON ECM.	All functions inhibited except platform down.
P104:	P104: LCON DRIVER 4 FAULT	LCON driver 4 failed.	• LCON ECM.	All functions inhibited except platform down.
P105:	P105: LCON DRIVER 5 FAULT	LCON driver 5 failed.	• LCON ECM.	All functions inhibited except platform down.
P106:	P106:LCON COIL BRAKE/LC:Bat+	Open or short circuit of the brake circuit to battery positive.	• Open or short on driver load. • Faulty connections or harness. • LCON ECM.	All functions inhibited except platform down.
P107:	P107: LCON COIL FAULT MC:OPEN	Open or short circuit of the motor controller.	• Open or short on driver load. • Faulty connections or harness. • LCON ECM.	All functions inhibited except platform down.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
U001:	U001:SWITCHFAULT GCON MAIN FTN EN	Short circuit of the GCON main function enable switch at system startup.	• Short circuit of the GCON main function enable switch. • GCON ECM.	All GCON functions inhibited.
U002:	U002:SWITCHFAULT GCON PLAT UP	Short circuit of the GCON up directional switch at system startup.	• Short circuit of the GCON up directional switch. • GCON ECM.	All GCON functions inhibited except platform down.
U003:	U003:SWITCHFAULT GCON PLAT DOWN	Short circuit of the GCON down directional switch at system startup.	• Short circuit of the GCON down directional switch. • GCON ECM.	All GCON functions inhibited except platform up.
U004:	U004:SWITCHFAULT GCON LCD UP	Short circuit of the GCON LCD scroll up switch at system startup.	• Short circuit of the GCON LCD scroll up switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U005:	U005:SWITCHFAULT GCON LCD DOWN	Short circuit of the GCON LCD scroll down switch at system startup.	• Short circuit of the GCON LCD scroll down switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U006:	U006:SWITCHFAULT GCON LCD ENTER	Short circuit of the GCON LCD enter switch at system startup.	• Short circuit of the GCON LCD enter switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U007:	U007:SWITCHFAULT GCON LCD ESCAPE	Short circuit of the GCON LCD escape switch at system startup.	• Short circuit of the GCON LCD escape switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U014:	U014:SWITCHFAULT PCON DRIVE EN	Short circuit of the PCON drive enable switch at system startup.	• Short circuit of the PCON drive enable switch. • GCON ECM.	All PCON drive and steer functions inhibited.
U015:	U015:SWITCHFAULT PCON STEER LEFT	Short circuit of the PCON steer left switch at system startup.	• Short circuit of the PCON steer left switch. • GCON ECM.	All PCON drive and steer functions inhibited.
U016:	U016:SWITCHFAULT PCON STEER RIGHT	Short circuit of the PCON steer right switch at system startup.	• Short circuit of the PCON steer right switch. • GCON ECM.	All PCON drive and steer functions inhibited.
U017:	U017:SWITCHFAULT PCON HORN	Short circuit of the PCON horn switch at system startup.	• Short circuit of the PCON horn switch. • GCON ECM.	PCON horn switch function inhibited.
U018:	U018:SWITCHFAULT PCON LO DRIV SPD	Short circuit of the PCON low drive speed switch at system startup.	• Short circuit of the PCON low drive speed switch. • GCON ECM.	The machine is limited to low drive speed.
U019:	U019:SWITCHFAULT PCON LO LIFT SPD	Short circuit of the PCON low lift speed switch at system startup.	• Short circuit of the PCON low lift speed switch. • GCON ECM.	PCON platform up and down functions inhibited.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
U020:	U020:SWITCHFAULT PCON HI LIFT SPD	Short circuit of the PCON high lift speed switch at system startup.	• Short circuit of the PCON hi lift speed switch. • GCON ECM.	PCON platform up and down functions inhibited.
U021:	U021:SWITCHFAULT PCON UP	Short circuit of the PCON up directional switch at system startup.	• Short circuit of the PCON up directional switch. • GCON ECM.	PCON platform up function inhibited.
U022:	U022:SWITCHFAULT PCON DOWN	Short circuit of the PCON down directional switch at system startup.	• Short circuit of the PCON down directional switch. • GCON ECM.	PCON platform down functions inhibited.
U023:	U023:SWITCHFAULT PCON O/R ENABLE	Short circuit of the PCON outrigger enable switch at system startup.	• Short circuit of the PCON outrigger enable switch. • GCON ECM.	All outrigger functions inhibited.
U033:	U033:JSTICKFAULT OUT OF CAL RANGE	PCON drive joystick signal is outside acceptable calibration range at system startup.	• PCON drive joystick is not in neutral position at startup. • PCON joystick. • GCON ECM.	All PCON drive and steer functions inhibited.
U034:	U034:JSTICKFAULT OUT OF RANGE:HI	Short circuit of the PCON drive joystick signal to battery positive at system startup.	• Short circuit of the PCON drive joystick signal circuit. • PCON joystick. • GCON ECM.	All PCON drive and steer functions inhibited.
U035:	U035:JSTICKFAULT OUT OF RANGE:LO	Short circuit of the PCON drive joystick signal to battery negative at system startup.	• Short circuit of the PCON drive joystick signal circuit. • PCON joystick. • GCON ECM.	All PCON drive and steer functions inhibited.
U036:	U036:SWITCHFAULT GCON + PCON:ON	Mis-wiring or short circuit of GCON key switch.	• GCON key switch. • GCON ECM.	All functions inhibited.
U037:	U037:SWITCHFAULT FOOTSW PRESSED	Foot switch pressed at machine startup.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.
U038:	U038:SWITCHFAULT FOOTSWITCH:BAT+	Mis-wiring or short circuit of foot switch to battery positive.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.
U039:	U039:SWITCHFAULT FOOTSW:OPEN/BAT-	Mis-wiring, open or short circuit of foot switch to battery negative.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.
U040:	U040:SWITCHFAULT FOOTSW:TIMEOUT	PCON deck switch pressed at machine startup.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.

Type "UXXX" Faults

Type "UXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
U041:	U041:SWITCHFAULT GCON DECK ENABLE	GCON deck enable switch stuck closed, or depressed at system startup.	• GCON deck enable switch stuck closed. • GCON deck enable switch depressed at system startup. • GCON ECM.	All functions inhibited.
U042:	U042:SWITCHFAULT PCON DRIVE MODE	PCON drive mode switch stuck closed, or depressed at system startup.	• PCON drive mode switch stuck closed. • PCON drive mode switch depressed at system startup. • PCON ECM.	Drive functions inhibited.
U043:	U043:SWITCHFAULT PCON OR MODE	PCON outrigger mode switch stuck closed, or depressed at system startup.	• PCON outrigger mode switch stuck closed. • PCON outrigger mode switch depressed at system startup. • PCON ECM.	Outrigger extend/retract functions inhibited.
U045:	U045:SWITCHFAULT PCON FUNCTION EN	PCON function enable switch stuck closed, or depressed at system startup.	• PCON function enable switch stuck closed. • PCON function enable switch depressed at system startup. • PCON ECM.	All drive and steer functions inhibited.
U046:	U046:SWITCHFAULT PCON LIFT MODE	PCON lift mode switch stuck closed, or depressed at system startup.	• PCON lift mode switch stuck closed. • PCON lift mode switch depressed at system startup. • PCON ECM.	All functions operate except platform up/down.
U047:	U047:SWITCHFAULT PCON INDOOR MODE	PCON Switch Fault	• Indoor button pressed at system power up • Internally shorted switch on PCON indoor button	All functions operate except platform up/down.
U048:	U048:SWITCHFAULT PCON OUTDOOR MODE	PCON Switch Fault	• Outdoor button pressed at system power up • Internally shorted switch on PCON outdoor button	All functions operate except platform up/down.
U049:	U049:SWITCHFAULT BRAKE RELEASE	Brake Release Switch Fault	• Brake Release switch activated at system power up • Internally shorted brake release switch	All functions inhibited.

Type "FXXX" Faults

Type "FXXX" Faults

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
F001:	F001:SWITCHFAULT UP LIMIT1:BAT+	Short circuit of the up limit #1 switch at system startup.	• Short circuit of the up limit switch circuit. • Up limit #1 switch short circuit. • GCON ECM.	Platform up function inhibited.
F003:	F003:SWITCHFAULT DOWN LIMIT1:BAT+	Short circuit of the down limit #1 switch at system startup.	• Short circuit of the down limit switch circuit. • Down limit #1 switch short circuit. • GCON ECM.	All functions inhibited except platform down.
F005:	F005:SWITCHFAULT POTHOLE:BAT+	Short circuit of the pothole limit #1 switch at system startup.	• Short circuit of the pothole switch circuit. • Pothole limit #1 switch short circuit. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.
F007:	F007:SWITCHFAULT CHASSISTILT:BAT+	Short circuit of the chassis digital tilt switch at system startup.	• Short circuit of the chassis digital tilt switch circuit. • Chassis digital tilt switch short circuit. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.
F008:	F008:SENSORFAULT OVLD XDUCER:BAT+	Short circuit of the Platform Overload Transducer circuit to battery positive.	• Short circuit in the transducer circuit. • Faulty pressure transducer. • GCON ECM.	All functions inhibited.
F009:	F009:SENSORFAULT OVLD XDUCER:BAT-	Short circuit of the Platform Overload Transducer circuit to battery negative.	• Short circuit in the transducer circuit. • Faulty pressure transducer. • GCON ECM.	All functions inhibited.
F010:	F010:SENSORFAULT PLAT HEIGHT:BAT+	Short circuit of the Platform Height Sensor circuit to battery positive.	• Short circuit in the platform height circuit. • Faulty platform height sensor. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.
F011:	F011:SENSORFAULT PLAT HEIGHT:BAT-	Short circuit of the Platform Height Sensor circuit to battery negative.	• Short circuit in the platform height circuit. • Faulty platform height sensor. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.

Type "FXXX" Faults

Type "FXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
F012:	F012:SENSORFAULT LEVEL PITCH:BAT+	Short circuit of the Level Pitch Sensor circuit to battery positive.	• Short circuit in the level pitch sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F013:	F013:SENSORFAULT LEVEL PITCH:BAT-	Short circuit of the Level Pitch Sensor circuit to battery negative.	• Short circuit in the level pitch sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F014:	F014:SENSORFAULT LEVEL ROLL:BAT+	Short circuit of the Level Roll Sensor circuit to battery positive.	• Short circuit in the level roll sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F015:	F015:SENSORFAULT LEVEL ROLL:BAT-	Short circuit of the Level Roll Sensor circuit to battery negative.	• Short circuit in the level roll sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F016:	F016:SENSORFAULT LF RIGGER:BAT+	Short circuit of the Left Front Outrigger Sensor circuit to battery positive.	• Short circuit in the left front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F017:	F017:SENSORFAULT LF RIGGER:BAT-	Short circuit of the Left Front Outrigger Sensor circuit to battery negative.	• Short circuit in the left front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F018:	F018:SENSORFAULT RF RIGGER:BAT+	Short circuit of the Right Front Outrigger Sensor circuit to battery positive.	• Short circuit in the right front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Right front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F019:	F019:SENSORFAULT RF RIGGER:BAT-	Short circuit of the Right Front Outrigger Sensor circuit to battery negative.	• Short circuit in the right front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Right front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F020:	F020:SENSORFAULT LR RIGGER:BAT+	Short circuit of the Left Rear Outrigger Sensor circuit to battery positive.	• Short circuit in the left rear outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left rear outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F021:	F021:SENSORFAULT LR RIGGER:BAT-	Short circuit of the Left Rear Outrigger Sensor circuit to battery negative.	• Short circuit in the left rear outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left rear outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F022:	F022:SENSORFAULT RR RIGGER:BAT+	Short circuit of the Right Rear Outrigger Sensor circuit to battery positive.	• Short circuit in the right rear outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Right rear outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.

Type "FXXX" Faults

Type "FXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
F023:	F023:SENSORFAULT RR RIGGER:BAT-	Short circuit of the Right Rear Outrigger Sensor circuit to battery negative.	- Short circuit in the right rear outrigger sensor circuit. - Faulty pressure transducer. - GCON ECM.	Right rear outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F034:	F034:SENSORFAULT OVLD:PLAT HEIGHT	Height Sensor settings are not calibrated correctly.	- Height sensor needs calibration for stowed height and maximum height settings. - GCON or PCON ECM.	All functions inhibited.
F035:	F035:SENSORFAULT OVLD RANGE:LOW	Height Sensor setting is lower than calibrated stowed height.	- Height sensor not calibrated for stowed height correctly. - GCON or PCON ECM.	All functions inhibited.
F036:	F036:SENSORFAULT OVLD RANGE:HI	Height Sensor setting is higher than calibrated maximum height.	- Height sensor not calibrated for maximum height correctly. - GCON or PCON ECM.	All functions inhibited.
F072:	F072:SENSORFAULT FUNCTION CUT B+	Function Cutout sensor B+ fault.	- Faulty extension deck limit switch. - Faulty gate proximity switch. - GCON ECM.	All functions inhibited.
F073:	F073:SENSORFAULT DECK SENSOR B+	Power Deck sensor B+ fault.	- Short circuit in deck sensor input circuit. - GCON ECM.	All functions inhibited.
F077:	F077:BATTERY TEMPERATURE HIGH	Operating temperature has exceeded the max allowable discharge/charge temperature for the battery	High current draw and/or high ambient temperature have caused battery temperature to exceed maximum allowable functional temperatures	All functions inhibited.
F078:	F078:BATTERY TEMPERATURE LOW	Operating temperature is below the min allowable discharge/charge temperature for the battery	Low current draw and/or low ambient temperature have caused battery temperature to exceed minimum allowable functional temperatures	All functions inhibited.
F081:	F081:DCON ECM THERM PROTECTION	DCON thermal protection	- Controller is operating in an extreme environment. - Excessive load on vehicle. - Improper mounting of controller which is preventing controller cooling. - Controller is performance-limited at this temperature.	All functions inhibited except platform down.
F082:	F082:MOTOR RF ENCODER FAULT	Right front motor encoder fault	- Encoder Steps parameter does not match the actual motor encoder. - Verify parameter settings: AC Motor Setup >> Quadrature Encoder>> Encoder Steps. - Motor encoder failure. - Bad crimps or faulty wiring. - External load impedance on the +5V supply is too low.	All functions inhibited except platform down.
F083:	F083:MOTOR LF ENCODER FAULT	Left front motor encoder fault	- Encoder Steps parameter does not match the actual motor encoder. - Verify parameter settings: AC Motor Setup >> Quadrature Encoder>> Encoder Steps. - Motor encoder failure. - Bad crimps or faulty wiring. - External load impedance on the +5V supply is too low.	All functions inhibited except platform down.
F084:	F084:MOTOR RF STALL/ENCODER	Right front motor stall	- Stalled motor. - Motor encoder failure. - Bad crimps or faulty wiring. - Problems with power supply for the motor encoder.	All drive and steer functions inhibited.
F085:	F085:MOTOR LF STALL/ENCODER	Left front motor stall	- Stalled motor. - Motor encoder failure. - Bad crimps or faulty wiring. - Problems with power supply for the motor encoder.	All drive and steer functions inhibited.
F086:	F086:RH BRAKE FAIL SET	Right emergency brake set failed	- Vehicle movement sensed after EM Brake has been commanded to set. - EM Brake will not hold the motor from rotating.	All functions inhibited except platform down.
F087:	F087:MOTOR LIMIT RUN	Motor limit run	- Limited Operating Strategy (LOS) control mode has been activated. - Motor encoder failure. - Bad crimps or faulty wiring. - Vehicle is stalled.	All drive and steer functions inhibited.
F088:	F088:LH BRAKE FAIL SET	Left emergency brake set failed	- Vehicle movement sensed after EM Brake has been commanded to set. - EM Brake will not hold the motor from rotating.	All functions inhibited except platform down.
F129:	F129:TILT SENSOR FAULT	UTS mismatch fault. Two accelerometers inside UTS mismatch.(universal tilt sensor)	Faulty UTS.	All functions inhibited.

Type "FXXX" Faults

Type "FXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
F130:	F130:LIFT MOTOR ENCODER FAULT	Lift Motor Encoder fault.	- Encoder Steps parameter does not match the actual motor encoder. - Motor encoder failure. - Bad crimps or faulty wiring.	All functions inhibited except platform down.
F131:	F131:LCON ECM THERM PROTECTION	LCON Thermal Protection.	- Controller is operating in an extreme environment. - Excessive load on vehicle. - Improper mounting of controller which is preventing controller cooling. - Controller is performance-limited at this temperature.	All functions inhibited except platform down.
F132:	F132:LIFT MOTOR TEMP CUTBACK	Lift Motor Temp Cutback.	DCON ECM.	All functions inhibited except platform down.
F133:	F133:LIFT MOTOR TEMP SENSOR	Lift Motor Temp Sensor.	- Short circuit in the platform height circuit. - Faulty lift motor sensor. - GCON ECM.	All functions inhibited except platform down.
F134:	F134:LIFT MOTOR STALL/ENCODER	Lift Motor Stall.	- Stalled motor. - Motor encoder failure. - Bad crimps or faulty wiring. - Problems with power supply for the motor encoder.	All drive and steer functions inhibited.
F135:	F135:LIFT BRAKE FAIL SET	Lift EM Brake Set.	- Vehicle movement sensed after EM Brake has been commanded to set. - EM Brake will not hold the motor from rotating.	All functions inhibited except platform down.

Type "CXXX" Faults

Type "CXXX" Faults

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
C001:	C001:GCON ECM FAULT TYPE 1	GCON ECM CRC check error.	- Incorrect software file. - GCON ECM internal failure.	All functions inhibited.
C004:	C004:GCON ECM FAULT TYPE 4	GCON ECM master switch error.	- Short circuit in the master switch circuit. - GCON ECM.	All functions inhibited.
C005:	C005:GCON ECM FAULT TYPE 5	GCON ECM safety switch error.	- Short circuit in the safety switch circuit. - GCON ECM.	All functions inhibited.
C006:	C006:GCON ECM FAULT TYPE 6	GCON input redundancy error.	- Input conditioning circuit failure. - GCON ECM.	All functions inhibited.
C007:	C007:GCON ECM FAULT TYPE 7	GCON ECM inter-processor communication error.	- Incorrectly programmed device. - Error in loading software on device. - GCON ECM.	All functions inhibited.
C009:	C009:GCON ECM FAULT TYPE 9	GCON ECM fault type 9.	Contact Genie support.	All functions inhibited.
C010:	C010:SECONDARY NOT PROGRAMMED	GCON secondary processor not programmed.	- Incorrectly programmed device. - Error in loading software on device. - GCON ECM.	All functions inhibited.
C021:	C021:PCON NOT DETECTED	PCON not detected error.	- PCON disconnected. - CAN communication failure. - GCON or PCON ECM.	All functions inhibited.
C023:	C023:MACHINE MODEL FAULT	Discrepancy between model detected and model programmed.	- Incorrect machine model programmed. - GCON or PCON ECM.	All functions inhibited.
C025:	C025:SYSTEMFAULT PLAT OVLD:NOCAL	Platform overload system not calibrated.	- Platform overload system not calibrated. - GCON or PCON ECM.	All functions inhibited.
C028:	C028:SERVICE OVERRIDE MODE ON	Machine is in service override mode.	Machine programmed for use in service override mode.	All functions inhibited except for down function and up function. Platform can be elevated only once with the maximum elevate time of XX seconds. Elevate time XX depends on machine model.

Type "CXXX" Faults

Type "CXXX" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
C029:	C029:SYSTEMFAULT O/R SENSOR:NOCAL	Outtrigger not calibrated.	• Outtrigger not calibrated. • GCON ECM.	All functions operate. Will not lift above the down limit switch.
C053:	C053:PCON-GCOM SOFTWARE MISMATCH	Software revisions do not match between the PCON and GCON.	• Short circuit in the safety switch circuit. • GCON ECM.	All functions inhibited.
C055:	C055:DCON ECM Fault Type 01	Hardware failure of drive controller.	• DCON ECM.	Performance reduced or all functions inhibited.
C056:	C056:DCON ECM Fault Type 02	Setup initialization failure of the drive controller at system startup.	• Drive input active at system startup. • Faulty drive joystick. • Incorrect software.D14 • DCON ECM.	Performance reduced or all functions inhibited.
C058:	C058:DCON ECM Fault Type 04	Drive controller voltage out of range.	• Battery charger connected to AC power source. • Batteries to low. • DCON ECM.	All functions inhibited.
C059:	C059:DCON ECM Fault Type 05	Capacitor charge failure of the drive controller.	• Open on B+ or B- to the drive controller. • DCON ECM.	All functions inhibited.
C061:	C061:DCON ECM Fault Type 07	Open circuit of the key switch circuit or battery positive / negative of the drive controller.	• Open circuit of the motor controller harness. • Open circuit to B+ and/or B-. • Key switch relay power not closed. • DCON or GCON ECM.	All functions inhibited.
C063:	C063:DCON ECM Fault Type 09	Communication error of the CAN circuit between the GCON and the motor controller.	• Open circuit of the motor controller harness. • DCON or GCON ECM.	All functions inhibited.
C064:	C064:DCON ECM Fault Type 10	Output error of the thermal sensor circuit of the motor controller.	• DCON ECM.	All functions inhibited.
C077:	C077:DISABLED BY OWNER	Telematics Remote Disable.	• Machine disabled.	All functions inhibited.
C078:	C078:TELEMATICS NOT DETECTED	Telematics device not detected by GCON.	• Telematics device not connected. • Telematics device not installed.	Telematics is installed: Connect the telematics device to the GCON. Telematics is not installed: Navigate to the options menu on the machine and disable telematics.
C079:	C079:MOTOR SPEED MONITOR	Motor speed supervision.	• Motor speed detected exceeding the limit set by the Max Speed Supervision parameter. • Mis-adjusted Max Speed Supervision parameters.	All functions inhibited except platform down.
C080:	C080:MOTOR TRAVEL MONITOR	Motor travel supervision.	• With the vehicle in the stopped state, the motor frequency and/or phase current detected exceeding the limit set by the Travel Control Supervision parameter. • Mis-adjusted Travel Control Supervision parameters.	All functions inhibited except platform down.
C081:	C081: MOTOR INTERLOCK FAULT	Motor interlock monitor.	• During an interlock braking event, the motor speed exceeded the limit set by the Interlock Braking Supervision parameters.	All functions inhibited except platform down.
C082:	C082:SYSTEMFAULT OHL:NoCal	Outdoor height limit not calibrated.	• OHL not Calibrated.	All functions inhibited except platform down.
C083:	C083:DCON ECM SETUP INVALID	DCON startup configuration failed.	• Parameter is not modified successfully by GCON.	All drive and steer functions inhibited.
C084:	C084:BATT MGNT SYS NOT DETECTED	Battery not detected.	• Battery management system	All functions inhibited.
C085:	C085:BATTERY END OF LIFE	Battery has reached end of life.	• Batteries primary internal fuse has been blown • Battery has reached 0% State of Health	All functions inhibited.
C086:	C086:BMS FAULT	Battery management system internal error.	• Battery management system	All functions inhibited.

Note: An error code C053 will be displayed if an updated PCON is installed on a machine that has not had the GCON updated. The GCON must be updated, or the original PCON must be installed back on the machine.

Type "CXXX" Faults

Type "E-Lift" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
C087:	C087: LCON NOT DETECTED	LCON Not Detected.	• LCON disconnected. • CAN communication failed.	Lifting/Lowering disabled.
C088:	C088: LCM NOT DETECTED	LCM Not Detected.	• Load cell module disconnected. • CAN communication failed.	All functions inhibited except platform down.
C089:	C089: LCON ECM FAULT TYPE 01	LCON ECM Fault: Hardware.	• Failure to read or write to nonvolatile (NVM) memory. • Internal controller fault detected. • Physical damage from external sources/events. • Battery overcharged. • Excessive motor or controller heating. • Firmware in the controller is incorrect. • CRC of the application or OS does not match. • Application built with an incompatible OS version. • Autobaud failure on linked controller CAN port. • Internal error with the controller. • Software loaded (.cdev) is not compatible with the controller hardware. • Mistuned Hazardous_Direction_Response_Time parameter. • Mistuned Hazardous_Accel parameter. • Mistuned Hazardous_Speed_Error parameter. • Mistuned Hazardous_Throttle_Response_Time parameter • Vehicle is moving when calibrating.	Lifting/Lowering disabled.
C090:	C090: LCON ECM FAULT TYPE 02	LCON ECM Fault: ECM Setup.	• Motor setup is required. • Incorrect sequence in application of Keyswitch, Interlock, Direction, or Throttle. • Faulty wiring, crimps or switches at KSI, interlock, Direction, or Throttle. • Moisture in above-noted digital input switches causing invalid (real) On/Off state. • Changing a safety-based parameter when interlock is ON. • Internal controller fault. • PDO Map has too many data bytes assigned or has objects mapped that are not compatible. • Parameter value detected outside of the limits. • Incorrect position feedback type choosen for motor technology in use. • Dual drive is enabled in torque mode. • Dual drive enabled on only one controller. • A Driver Output is used for two or more functions. • Mistuned Dual Drive CAN parameters. • Conflicting CAN Node Ids for Dual Drive.	Lifting/Lowering disabled.

Type "CXXX" Faults

Type "E-Lift" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
C091:	C091: LCON ECM FAULT TYPE 03	LCON ECM Fault: ECM Valve.	• Not defined	
C092:	C092: LCON ECM FAULT TYPE 04	LCON ECM Fault: Over Under V.	• Non-controller system drain on battery/keyswitch circuit wiring. • Resistance in low power (KSI) circuit is too high. • KSI disconnected while driving. • Battery parameters are misadjusted. • Battery resistance too high for given regen current. • Battery disconnected while regen braking. • Battery voltage applied to KSI (pin1) exceeds the severe overvoltage limit. • Regen braking currents elevated the battery voltage	Lifting/Lowering disabled
C093:	C093: LCON ECM FAULT TYPE 05	LCON ECM Fault: Cap Charge.	• An external load on the capacitor bank (B+ connection terminal) that prevents the capacitor bank from charging.	Lifting/Lowering disabled
C094:	C094: LCON ECM FAULT TYPE 06	LCON ECM Fault: PEV Disconnect.	• Not defined.	
C095:	C095: LCON ECM FAULT TYPE 07	LCON ECM Fault: ECM Key Off.	• Not defined.	
C096:	C096: LCON ECM FAULT TYPE 09	LCON ECM Fault: LCON No CAN Msg.	• Not defined.	
C097:	C097: LCON ECM FAULT TYPE 10	LCON ECM Fault: LCON Temp Sensor.	• Controller is operating in an extreme environment. • Excessive load on vehicle. • Improper mounting of controller.	Lifting/Lowering disabled
C098:	C098: LCON MOTOR TRAVEL MONITOR	LCON Motor Travel Supervision.	• Motor frequency and/or phase current detected exceeding the limit set by the Travel Control Supervision parameter. • Misadjusted Travel Control Supervision parameters.	Lifting/Lowering disabled
C099:	C099: PCON NOT AUTHORIZED	PCON not Authorized.	• Handshake process failed. • Non-genuine PCON installed onto the machine.	All functions operate.
C100:	C100: DCON NOT AUTHORIZED	DCON not Authorized.	• Handshake process failed. • Non-genuine DCON installed onto the machine.	All functions operate.

Type "CXXX" Faults

Type "E-Lift" Faults, (continued)

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
C101:	C101: LCON NOT AUTHORIZED	LCON not Authorized.	- Handshake process failed. - Non-genuine LCON installed onto the machine.	All functions operate.
C102:	C102: TILT SENSOR NOT AUTHORIZED	Tilt Sensor not Authorized.	- Handshake process failed. - Non-genuine Tilt Sensor installed onto the machine.	All functions operate.
C103:	C103: LOAD SENSOR NOT AUTHORIZED	Load Cell not Authorized.	- Handshake process failed. - Non-genuine Load Cell installed onto the machine.	All functions operate.
C104:	C104: BATT CHRGR NOT AUTHORIZED	Battery Charger not Authorized.	- Handshake process failed. - Non-genuine Battery Charger installed onto the machine.	All functions operate.
C105:	C105: BATT MGMT NOT AUTHORIZED	Battery not Authorized.	- Handshake process failed. - Non-genuine Battery installed onto the machine.	All functions operate.
C106:	C106: DCON LCON COMMUNICATION	DCON LCON Communication.	- CAN communication failure. - DCON or LCON ECM.	Lifting/Lowering disabled
C107:	C107: LCON PDO TIMEOUT	LCON PDO timeout.	The time between CAN PDO messages received exceeded the PDO timeout period as defined by the Event Timer parameter.	Lifting/Lowering disabled
C108:	C108: LIFT MOTOR SPEED MONITOR	Lift Motor Speed Supervision.	- Motor speed detected exceeding the limit set by the Max Speed Supervision parameter. - Misadjusted Max Speed Supervision parameter.	Lifting/Lowering disabled
C109:	C109: LIFT MOTOR FEEDBACK	Lift Motor Encoder Failure.	- Limited Operating Strategy (LOS) controlmode has been activated; as a result of either an Encoder Fault (flash code 3-6) or a Stall Detected fault (flash code 7-3). - Motor Encoder failure. - Bad crimps of faulty wiring. - Vehicle is stalled.	Lifting/Lowering disabled
C110:	C110: MC SOFTWARE MISMATCH	MC Software Mismatch.	- Incompatible DCON installed onto the machine. - DCON ECM.	All functions inhibited.
C111:	C111: SYSTEMFAULT MAX HEIGHT:NoCal	Max Height Not Calibrated.	Max Height Calibration process not performed.	All functions inhibited except platform down.

Battery Charger

Battery Charger

The charger continuously monitors internal and external conditions. Fault and Error codes are generated by the charger to indicate that an internal or external malfunction has been detected in the electrical system. The types of Diagnostic Trouble Codes that may occur are explained below.

Type "F" codes – Indicate an internal fault condition has occurred and caused charging to stop.

Type "E" codes - Indicate an external fault condition has occurred and caused charging to stop.

Type "P" code - Indicates charger programming mode is active. This will occur when the charger profile is being configured.

Type "USB" code - Indicates the USB interface is active, and the USB should not be removed. This will occur when the charger firmware is being updated.



Solid red - Internal fault condition. Type "F" fault code.



Flashing amber - External fault condition. Type "E" error code.



Flashing green - USB interface is active.

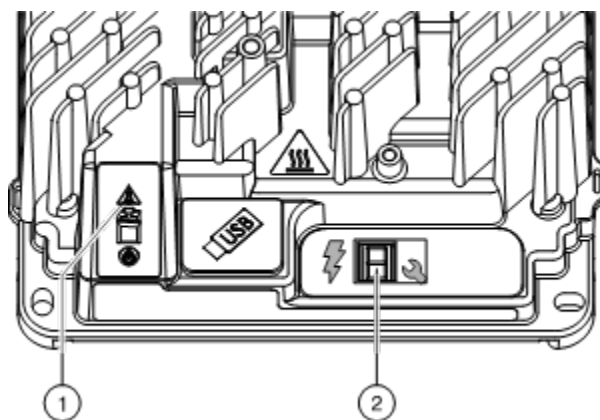


Solid green - Firmware update is complete. Remove USB from charger.

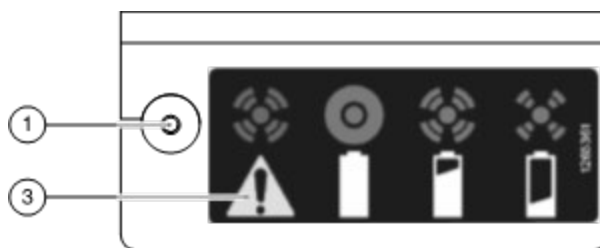
Codes are displayed as F, E or P and followed by three numbers and a period (example: E-0-0-4). The charger can display multiple codes.

If the batteries are changed from one type to another, (example: flooded lead acid to AGM), the charger profile must be programmed. Refer to Repair Procedure, *Programming the Charger Profile*.

Battery Charger



Charger status indicators



Remote charger status indicators

- 1 Error / Fault / USB indicator
- 2 Charger profile / Error display
- 3 Error / Fault

Charger Fault Codes

Charger Fault Code	Genie Fault ID	Genie Fault Type	Description	Possible Causes	Solution
F-0-0-1	135	8	DC-DC failure: LLC excessive leakage fault.	Internal charger fault.	Disconnect AC input and battery pack for 30 seconds. If fault does not clear, replace charger.
F-0-0-2	135	8	PFC failure: PFC excessive leakage fault.		
F-0-0-3	135	8	PFC taken too long to boost.		
F-0-0-4	135	8	Charger unable to calibrate current offset.		
F-0-0-5	135	8	Output relay voltage too high when closed.		

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-0-1	131	23	High battery voltage	Battery voltage too high to charge.	Check battery voltage and cable connections. Self clearing when condition has been corrected.
E-0-0-2	131	24	Low battery voltage detected prior to starting charge cycle	Battery voltage too low to charge.	Check battery voltage and cable connections. Check battery condition. Self clearing when condition has been corrected.
E-0-0-3	135	28	Charge timeout	Charge timeout caused by battery pack not reaching required voltage within safe time limit.	Charge output reduced due to high temperature. Charge at lower temperature. Check battery condition. Replace damaged battery. Check battery connections. Battery pack deep discharge. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.
E-0-0-4	204	21	Defective battery	Battery pack could not be trickle charged up to the minimum voltage.	Check battery condition. Replace damaged battery. Check battery connections. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.
E-0-0-7	264	23	Amp hour limit exceeded	Safety limit exceeded.	Check battery condition. Replace damaged battery. Check battery connections. Battery pack deep discharge. High parasitic loads on battery pack while charging. Disconnect parasitic loads. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-0-8	233	19	Battery temperature out of range	Battery temperature sensor.	Check temperature sensor and electrical connections. Rest charger. Disconnect AC power for 30 seconds. Self clearing when condition has been corrected.
E-0-1-1	135	8	Charger disabled by external command	Battery pack not being charged..	Power outage. AC power disconnected before charging complete.
E-0-1-2	135	8	Reverse polarity	Battery connected incorrectly.	Check battery connections. Self clearing when condition has been corrected.
E-0-1-3	204	21	Battery does not take current	Battery voltage is detected, but charger is unable to charge the battery pack..	Incorrect profile for battery type. Installed from a different machine with different battery type. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.
E-0-1-6	135	8	Software update failed	Software update failed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-1-7	265	21	USB error	USB not connected correctly.	Remove and re-insert USB Flash Drive Disconnect AC input and battery pack for 30 seconds.
E-0-1-8	135	8	Slot CRC error	Software update failed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-1-9	135	29	Hardware does not support software	Charger hardware does not support software version being programmed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-2-0	266	31	No algorithm selected	Charger profile not programmed.	Program the charger for the proper battery profile.

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-2-1	254	23	High battery voltage detected while charging	Battery voltage too high as detected by charger profile.	Check battery voltage and cable connections. Confirm charger profile is programmed for correct battery type. Self clearing when condition has been corrected.
E-0-2-2	254	24	Low battery voltage detected while charging	Battery voltage too low as detected by charger profile.	Check battery voltage and cable connections. Confirm charger profile is programmed for correct battery type. Self clearing when condition has been corrected.
E-0-2-3	267	23	High AC voltage	AC voltage greater than 270 VAC.	Connect to a stable AC source between 85 - 270 VAC / 45 - 65 Hz. Self clearing when condition has been corrected.
E-0-2-4	135	8	Failure to initialize	Charger failed to turn on correctly.	Disconnect AC input and battery pack for 30 seconds. Contact Genie Product Support.
E-0-2-5	267	24	Low AC voltage oscillation	AC voltage unstable.	Connect to a stable AC source between 85 - 270 VAC / 45 - 65 Hz. Using an undersized generator. Self clearing when condition has been corrected.
E-0-2-6	135	8	Script failure	Software update failed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-2-7	265	21	USB over current fault	USB over current protection has been tripped.	Remove and re-insert USB Flash Drive Use different USB Flash drive.
E-0-2-8	266	21	Charge profile incompatibility	Selected charger profile is not compatible with charger software.	Update charger software. Confirm charger profile is programmed for correctly. Updating current firmware and error occurs, contact Genie Product Support.

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-2-9	135	18	CAN Bus error	CAN Bus network error.	Check CAN Bus connections. CAN modules not functioning correctly. Termination resistance is ~60 ohms.
E-0-3-0	135	8	COMM battery module error	CAN Bus battery module error.	CAN Buss battery module not functioning correctly.
E-0-3-1	135	8	Vref for ADC measurements triggered alarm	Internal charger error.	Disconnect AC input and battery pack for 30 seconds. Contact Genie Product Support.
E-0-3-2	135	13	CAN Bus heartbeat error	CAN Bus heartbeat error.	CAN Bus devices not functioning correctly.
E-0-3-6	233	21	Battery temperature sensor charge profile	Battery temperature sensor is required by charger profile but is not installed.	Check battery temperature sensor connections. Check charger profile and confirm battery temperature sensor is installed.
E-0-3-7	135	8	CAN Open Error	CAN Open reprogramming failed.	Download CAN Open software. Reprogram charger using USB Flash Drive.
E-0-3-8	268	21	Stalled cooling fan	Fan obstructed.	Inspect fan for damage. Clear blockage.

Lithium Battery - Option

90AH Lithium Battery - Option

The battery is a modular battery system featuring an Intelligent Battery Management System (BMS) that offers safety protection for over/under voltage, overcharge/discharge current, short circuit and over/under temperature conditions.

Proper battery condition is essential to good machine performance and operational safety. Damaged cables and connections can result in component damage and hazardous conditions.

Note: Installation and maintenance shall be completed by a person trained and qualified to service lithium-ion batteries.

⚠ WARNING

Electrocution/burn hazard. Contact with electrically charged circuits could result in death or serious injury. Remove all rings, watches and other jewelry.

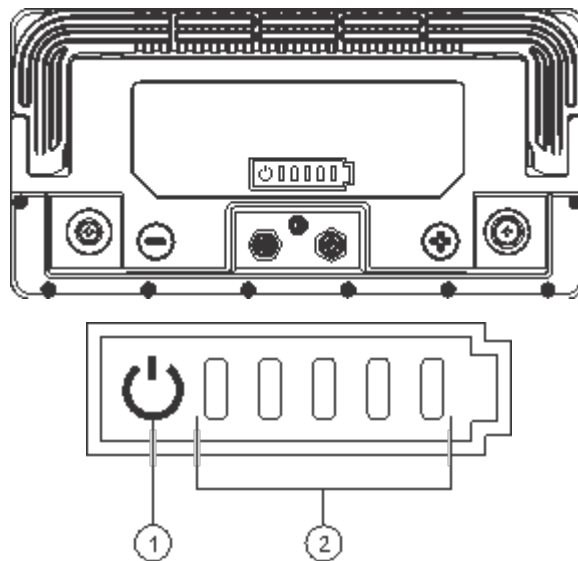
Battery manufacture recommends that the battery be placed in Ship Mode when the machine is stored for an extended period of time to extend battery storage life. When in Ship Mode the machine is inoperable until Wake Up Mode is activated.

SOC - State of Charge

Ship Mode and Wake Up Mode

Ship mode: Press and hold the SOC button for 20 seconds.

Wake Up mode: Press and hold the SOC button for 5 seconds.



- 1 SOC button
- 2 SOC / Fault indicator LED's

Lithium Battery - Option

Battery SOC Indicators

Press and release the SOC button.

- ⦿ Result: If the battery is above 10% SOC, the green indicators will remain on for 5 seconds.
- ⦿ Result: If the battery is below 10% SOC, the red indicator will flash for 10 seconds.

Battery SOC	LED Status
81 - 100%	
61 - 80%	
41 - 60%	
21 - 40%	
11 - 20%	
<10%	

Lithium Battery - Option

Battery Fault Codes

Fault ID	LED Status	Description	Solution
F0		Over temperature (cells)	Wait for temperature to drop into acceptable operating range
F1		Over temperature (BMS)	Wait for temperature to drop into acceptable operating range
F2		Under temperature (charge)	Wait for temperature to drop into acceptable operating range
F3		Over current (recoverable)	If charging, power cycle charger. Replace charger if problem persists. If discharging, battery will resume operation after discharge has been stopped.
F4		Over current (permanent fault)	Disconnect and replace battery.
F5		Short circuit	Check all connection points to battery system.
F6		Cell under voltage during discharge only	Recharge battery.
F7		Cell over voltage during charge only (primary)	Discharge battery down to 20% SOC, then recharge battery.

Lithium Battery - Option

Battery Fault Codes

Fault ID	LED Status	Description	Solution
F8		Cell over voltage during charge only (secondary)	Disconnect and replace battery.
F9		Safety under voltage	Recharge battery.
F10		Cell pre-charge fault	Power cycle charger. Replace charger if problem persists.
F11		Charge fault	Power cycle charger. Replace charger if problem persists.
F12		Under temperature (discharge)	Wait for temperature to rise into acceptable operational range.
F13		Miscellaneous fault	Power cycle battery. Replace battery if problem persists.
F14		Pre-discharge	Enable the battery without load and check if fault goes away. If fault is not present, reconnect to system. If fault persists, disconnect and replace battery.
F15		Permanent fault	Disconnect and replace battery.